

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1706

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes

Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Ananya S (192311099) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Ananya S

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand
---------	----	--	----------------------------------	-------------------------------------	---

ANSWERS

1. SMART HOME AUTOMATION SYSTEM

Introduction

A Smart Home Automation System is an AI-based system that automatically controls lighting, temperature, and security by using real-time sensor data such as motion detection, temperature, and time of day. It can also learn user preferences over time to improve comfort, energy efficiency, and security.

Types of Intelligent Agents 1.

Simple Reflex Agent

Working:

A Simple Reflex Agent acts only on the **current input (percept)** using predefined **if-then rules**. It does not remember previous events.

Example in Smart Home

- If motion is detected → Turn on the lights.
- If no motion is detected → Turn off the lights.

- If temperature is above 30°C → Turn on the air conditioner.

Advantages

- Fast response.
- Easy to implement.

Limitations

- Cannot learn user preferences.
- Cannot make decisions based on past experiences.

2. Model-Based Agent

Working

A Model-Based Agent keeps an **internal model (memory)** of the environment and uses both current and past information to make decisions.

Example in Smart Home

- Remembers that the bedroom window is open.
- If the AC is ON and the window is open, it alerts the user or switches off the AC to save energy.

Advantages

- Works in partially observable environments.
- Makes better decisions using stored information.

Limitations

- More complex than a Simple Reflex Agent.

3. Goal-Based Agent

Working

A Goal-Based Agent chooses actions that help achieve a specific goal.

Example in Smart Home Goals:

- Maintain room temperature at **24°C**.
- Keep the home secure.
- Ensure lights are available when needed.

The agent selects actions that help achieve these goals.

Advantages

- Can plan actions.
- Produces better decisions than reflex agents.

Limitations

- Requires planning, which takes more time.

4. Utility-Based Agent

Working

A Utility-Based Agent selects the action that provides the **maximum overall benefit (utility)** by considering multiple factors. **Example in Smart Home** The agent balances:

- User comfort
- Electricity consumption
- Security
- User preferences Example:

If electricity prices are high, the agent may increase the AC temperature slightly to reduce energy usage while keeping the room comfortable.

Advantages

- Makes the best overall decisions.
- Optimizes comfort, energy efficiency, and security.

Limitations

- More computationally complex.

Comparison of Intelligent Agents

Agent Type	Working	Example in Smart Home
Simple Reflex Agent	Uses current input only	Turns lights ON when motion is detected
Model-Based Agent	Uses memory of previous states	Remembers window status before operating the AC
Goal-Based Agent	Works to achieve predefined goals	Maintains desired room temperature and security
Utility-Based Agent	Chooses the action with the highest overall benefit	Balances comfort, energy saving, and security

Most Effective Approach

The best approach is a **Hybrid Agent** that combines:

- **Simple Reflex Agent** for immediate responses (lights, alarms).
- **Model-Based Agent** for remembering environmental conditions and user habits.
- **Goal-Based Agent** for achieving desired comfort and security goals.
- **Utility-Based Agent** for selecting the most beneficial action by balancing comfort, energy efficiency, and security.

Justification

A Hybrid Agent is the most suitable because it:

- Responds quickly to real-time events.

- Learns and remembers user preferences.
- Plans actions to achieve home automation goals.
- Optimizes energy consumption while maintaining user comfort.
- Enhances home security through intelligent decision-making.
- Adapts to changing environmental conditions.

2. ROBOT NAVIGATION IN AN UNKNOWN MAZE

Introduction

A robot is placed in an **unknown maze** and must find a path from the **Start** to the **Exit** without prior knowledge of the maze. The robot uses **search algorithms** to explore the maze and intelligent agents to make decisions while navigating.

Uniform Cost Search (UCS)

Working

Uniform Cost Search (UCS) expands the node with the **lowest cumulative path cost** first. It guarantees the **least-cost path** when all path costs are non-negative.

Example

- The robot calculates the total cost of every possible path.
- It always selects the path with the minimum cost until it reaches the exit.

Advantages

- Finds the **optimal (least-cost) path**.
- Works well when different paths have different costs.
- Complete and optimal.

Disadvantages

- Requires more memory.

- Can be slower for very large search spaces.

Complexity

- **Time Complexity:** $O(b^d)$
- **Space Complexity:** $O(b^d)$

where:

- **b** = branching factor
- **d** = depth of the optimal solution

Iterative Deepening Search (IDS)

Working

Iterative Deepening Search repeatedly performs **Depth-First Search (DFS)** with increasing depth limits until the goal is found.

Example

- Search up to depth 1.
- If the exit is not found, search up to depth 2.
- Continue increasing the depth until the exit is reached.

Advantages

- Uses very little memory.
- Complete like BFS.
- Suitable when the depth of the goal is unknown.

Disadvantages

- Repeats the search at each depth level.
- May take more time due to repeated exploration.

Complexity

- **Time Complexity:** $O(b^d)$
- **Space Complexity:** $O(bd)$

Comparison of UCS and IDS

Feature	Uniform Cost Search (UCS)	Iterative Deepening Search (IDS)
Search Method	Lowest path cost first	Repeated depth-limited DFS
Path Cost	Considers path cost	Ignores path cost
Optimal	Yes	Yes (when all step costs are equal)
Complete	Yes	Yes
Time Complexity	$O(b^d)$	$O(b^d)$
Space Complexity	$O(b^d)$	$O(bd)$

Intelligent Agents

The robot behaves as an **Intelligent Agent** because it:

- Perceives the environment using sensors.
- Processes information.
- Selects appropriate actions.
- Moves toward the goal.

Agent Design and Decision-Making

Simple Reflex Agent

- Uses only the current perception.

- Cannot remember previous locations.
- Not suitable for an unknown maze.

Model-Based Agent

- Maintains an internal map of explored paths.
- Avoids revisiting previously explored locations.
- Makes better navigation decisions.

Goal-Based Agent

- Plans actions to reach the exit.
- Chooses paths that lead toward the goal.

Utility-Based Agent

- Selects the action that minimizes total cost, time, and risk.
- Produces the best overall solution.

Best Combination

Agent Type: Model-Based Goal-Based Agent

Search Strategy: Uniform Cost Search (UCS)

Justification

This combination is most suitable because:

- The **Model-Based Agent** stores information about explored paths and obstacles.
- The **Goal-Based Agent** focuses on reaching the exit efficiently.
- **Uniform Cost Search (UCS)** always finds the **least-cost path**, making navigation efficient when movement costs vary.
- Together, they reduce unnecessary exploration and improve decision-making.

